

Summary

I am a Postdoctoral Researcher at CREATIS Biomedical Imaging Laboratory, INSA-Lyon, France, working on scientific machine learning for forward and inverse PDE problems, with applications in computational fluid dynamics and medical imaging. I completed my M.Tech from IIT Delhi and Ph.D. from IIT Madras under the supervision of Prof. Balaji Srinivasan. I have over 2 years of industry experience as a Data Scientist at Vunet Systems and almost 3 years of postdoctoral research experience across India, the United States, and France. I was also awarded the Best PhD Thesis Award in Data Science by IIT Madras.

Education

Ph.D. – Mechanical Engineering, IIT Madras, India (Jan 15 – Mar 21)
 Supervisor: Prof. Balaji Srinivasan (Mech Engg & Wadhvani School of Data Science & AI)
 Thesis: *Physics Enhanced Machine Learning Methods for Flow Simulation*.

M.Tech – Applied Mechanics, IIT Delhi, India (Jul 12 – Nov 14)

B.E. – Mechanical, IET-DAVV, India (Jul 07 – May 11)

Research Experience

Postdoc – CREATIS Biomedical Imaging Laboratory, INSA-Lyon, France (Nov 24 – Now)
 Rapid PINN-related algorithms; flow reconstruction with partial data; inverse problems.

Postdoc – State University of New York at Albany NY, USA (Jul 23 – May 24)
 ML-based tropical cyclone intensity estimation

Postdoc – Robert Bosch Center for Data Science and AI, IIT Madras, India (Jul 22 – Mar 23)
 PDE-constrained medical image segmentation.

Industry Experience

Data Scientist – Vunet Systems, Bangalore, India (Apr 20 – Jun 22)
 ML analytics for business/performance observability platforms

Publications (Selected)

1. **Dwivedi, V.**, Sigovan, M., & Sixou, B. (2026). Meta-SPINN: Meta-learned basis adaptation for parametric PDEs. *IOP-MLST*. doi:[10.1088/2632-2153/ae94e2](https://doi.org/10.1088/2632-2153/ae94e2).
2. **Dwivedi, V.**, Sixou, B., & Sigovan, M. (2025). Curriculum learning-driven PI-ELMs for fluid flow simulations. *Neurocomputing*. doi:[10.1016/j.neucom.2025.130924](https://doi.org/10.1016/j.neucom.2025.130924).
3. **Dwivedi, V.**, Srinivasan, B., & Krishnamurthi, G. (2024). Physics-informed contour selection for rapid image segmentation. *Scientific Reports*. doi:[10.1038/s41598-024-57281-x](https://doi.org/10.1038/s41598-024-57281-x).
4. **Dwivedi, V.**, Parashar, N., & Srinivasan, B. (2020). Distributed learning machines for solving forward and inverse PDE problems. *Neurocomputing*. doi:[10.1016/j.neucom.2020.09.006](https://doi.org/10.1016/j.neucom.2020.09.006).
5. **Dwivedi, V.**, & Srinivasan, B. (2019). Physics informed extreme learning machine (PIELM) – a rapid method for the numerical solution of PDEs. *Neurocomputing*. doi:[10.1016/j.neucom.2019.12.099](https://doi.org/10.1016/j.neucom.2019.12.099).

Conference Proceedings

- Srinivasan, A. G., **Dwivedi, V.**, & Srinivasan, B. (2025). Towards Sustainable Scientific Machine Learning: Fast and Interpretable PDE Solvers via RBF-PIELM. In: *Proc. 13th ACM IKDD'25*. doi:[10.1145/3799830.3799865](https://doi.org/10.1145/3799830.3799865).
- **Dwivedi, V.** (2024). Global versus Local: Evaluating AlexNet Architectures for Tropical Cyclone Intensity Estimation. In: *Proc. 27th ICPR' 24*. doi:[10.1007/978-3-031-78186-527](https://doi.org/10.1007/978-3-031-78186-527).

Presentations (Selected)

- Daniel, A, **Dwivedi, V.** & Srinivasan, B. (Dec 2025). AB-PIELMS: Adaptive-Basis Physics-Informed Extreme Learning Machines for Residual-Driven Domain Decomposition. **AI&PDE: ICLR 2026 Workshop on AI and Partial Differential Equations, Rio de Janeiro, Brazil.** [open review](#).
- **Dwivedi, V.**, Sigovan, M., & Sixou, B. (Dec 2025). Curriculum-Learning PIELMs for Hemodynamic Flows. **17th OPT Workshop @ NeurIPS, San Diego, USA.** [open review](#).
- **Dwivedi, V.**, Sigovan, M., & Sixou, B. (Dec 2025). Empirical-Bayes XTFC for Inverse Parameter Estimation. **17th OPT Workshop @ NeurIPS, San Diego, USA.** [open review](#).
- Srinivasan, A G, **Dwivedi, V.** & Srinivasan, B. (Dec 2025). Deep vs. Shallow: Benchmarking Physics-Informed Neural Architectures on the Biharmonic Equation. **8th ML4PS Workshop @ NeurIPS, San Diego, USA.** [PDF link](#).

Honors & Service

- **Best PhD Thesis Award in Data Science (2021)**, IIT Madras. [Youtube link](#).
- **Invited Lecture:** “Scientific Machine Learning for Fluid Mechanics”, IIT Kanpur (6 Nov 2023).
- **Keynote Speaker:** International Conference in Applied Mechanics, Machine Learning and Applied Computation (2022), NIT Raipur.
- **MOOC:** Short course on Data-Centric AI, NPTEL. [Workshop link](#).
- **Reviewer:** *Neurocomputing*, *Scientific Reports*, *Applied Mathematical Modelling*.
- **GATE 2011:** AIR 325/81175 (Mechanical Engineering).
- **Scholarships:** MHRD scholarship during Masters and PhD.

Technical Skills

Programming	Python, MATLAB; TensorFlow; NumPy/SciPy; scikit-learn; optimization & numerical linear algebra
Domains	Physics-/biophysics-informed learning, PDE-constrained optimization, CFD, medical imaging, inverse problems
Tools	Git, Linux; LaTeX; data visualization and scientific plotting

References

Prof. Balaji Srinivasan

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Prof. Ganapathy Krishnamurthi

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